MINOR PROJECT REPORT

FPGA-Based Network Intrusion Detection System

Using Quantized Neural Networks and the FINN Framework

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Academic Year 2025-26

Abstract

This project presents the design and implementation of an FPGA-accelerated Network Intrusion Detection System (NIDS) leveraging quantized neural networks. Traditional software-based intrusion detection systems suffer from high latency and inadequate throughput when faced with modern high-speed network traffic. By deploying a quantized Multi-Layer Perceptron (MLP) on an FPGA using the FINN (Fast Inference on Neural Networks) framework, we achieve hardware-level inference at microsecond latency.

The system was trained and evaluated on the NSL-KDD benchmark dataset using Brevitas, a PyTorch-based quantization-aware training library. Three quantization levels were explored: 2-bit, 4-bit, and 8-bit. Results demonstrate that the 4-bit model achieves 79.41% accuracy with an F1 score of 0.7924, requiring only 9.47 KB of on-chip memory. The 8-bit model achieves the highest accuracy at 81.26%, while the 2-bit model reduces memory usage by 75% with only a 3.75% accuracy penalty. The ONNX-exported model was compiled through the FINN pipeline to generate HLS-ready hardware description code targeting the PYNQ-Z2 FPGA board (Zynq-7020 SoC).

Keywords: FPGA, Quantized Neural Networks, NIDS, FINN, Brevitas, NSL-KDD, Dataflow Architecture, HLS, Edge AI

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1. Introduction

1.1 Background

The exponential growth of internet-connected devices and services has significantly increased the attack surface for cyber threats. Network Intrusion Detection Systems (NIDS) serve as a critical line of defense, monitoring network traffic and flagging malicious behavior in real-time. However, as network speeds scale to 10 Gbps and beyond, traditional CPU-based NIDS solutions such as Snort and Suricata struggle to maintain line-rate processing without packet drops or unacceptable latency.

Field Programmable Gate Arrays (FPGAs) offer a compelling alternative for high-performance, low-latency inference tasks. Their reconfigurable hardware fabric allows for massively parallel, custom dataflow architectures that can process network packets at hardware speed with predictable timing and minimal power consumption.

1.2 Motivation

The convergence of two trends motivates this project: first, the increasing sophistication and volume of cyber attacks requiring faster detection; and second, the maturation of quantization-aware training techniques that allow neural networks to be compressed to 2-8 bit precision without catastrophic accuracy loss. Together, these enable small, efficient neural networks to be deployed directly on FPGA fabric for real-time classification.

1.3 Problem Statement

The core problem addressed in this project is: How can a neural network-based intrusion detection system be deployed on FPGA hardware to achieve microsecond-level inference latency while maintaining acceptable classification accuracy on the NSL-KDD benchmark dataset?

1.4 Objectives

- Design and train a quantized MLP for binary network intrusion classification on the NSL-KDD dataset
- Investigate the accuracy-memory trade-off across 2-bit, 4-bit, and 8-bit quantization levels
- Export the trained model to ONNX format and compile it using the FINN framework
- Generate HLS C++ code targeting the PYNQ-Z2 FPGA board
- Estimate and report hardware resource utilization (LUTs, BRAMs, FFs, DSPs) and inference latency
- Demonstrate the viability of FPGA-based edge AI for cybersecurity applications

1.5 Scope and Limitations

This project covers the full software pipeline from data preprocessing through quantization-aware training, ONNX export, and FINN compilation. Due to hardware availability constraints, actual FPGA deployment was not performed; however, synthesis reports and HLS-generated resource estimates are presented. The dataset used is NSL-KDD which, while widely used in academic research, does not reflect the full complexity of modern network traffic.

2. Literature Review

2.1 Traditional Intrusion Detection Systems

Early NIDS relied on signature-based detection using pattern matching, as implemented in tools like Snort (Roesch, 1999). While effective against known attacks, these systems fail against novel or polymorphic threats. Anomaly-based detection using statistical methods and later machine learning approaches were proposed to address this limitation, at the cost of higher false positive rates.

2.2 Machine Learning for Network Intrusion Detection

The application of machine learning to NIDS has been extensively studied. Decision trees, SVMs, and Random Forests were among the first ML methods applied to the KDD and NSL-KDD datasets. Deep learning approaches, including MLPs and CNNs applied to 1D traffic feature vectors, have since demonstrated superior accuracy. However, the computational cost of deep neural network inference on CPUs or GPUs makes real-time deployment at line rate challenging.

2.3 Quantized Neural Networks

Quantization reduces the precision of neural network weights and activations from 32-bit floating point to lower bit-widths (8, 4, 2, or 1 bit). Binary Neural Networks (BNNs), where weights and activations are constrained to +1/-1, were introduced by Courbariaux et al. (2016) and demonstrated that heavily quantized networks could still achieve competitive accuracy. The key insight is that for hardware deployment, the reduction in arithmetic complexity far outweighs the marginal accuracy loss from quantization.

2.4 FPGA Acceleration for Neural Networks

FPGAs have been used as neural network accelerators since the work of Farabet et al. (2009). Systolic array architectures and dataflow designs have both been explored. The dataflow approach, where each neural network layer maps to a dedicated hardware block with streaming interfaces, is particularly well-suited to FPGAs due to its predictable timing and elimination of off-chip memory bottlenecks.

2.5 The FINN Framework

FINN (Fast Inference on Neural Networks) was introduced by Umuroglu et al. (2017) at Xilinx Research Labs. It provides an end-to-end compilation pipeline from quantized PyTorch models (trained with Brevitas) through ONNX, FINN graph transformations, HLS C++ code generation, and finally Vivado synthesis to FPGA bitstream. The framework is specifically designed for deeply quantized networks and generates dataflow architectures where weights are stored entirely on-chip in BRAMs, eliminating DRAM bandwidth as a bottleneck.

3. System Design and Methodology

3.1 Overall System Architecture

The proposed system consists of three major components: a software preprocessing pipeline, a quantized neural network inference engine compiled to FPGA hardware, and an output classification stage. Network traffic features are extracted and normalized on a host processor, then streamed to the FPGA accelerator via an AXI-Stream interface. The FPGA runs the FINN-generated dataflow MLP and returns classification results in real-time.

3.2 Dataset — NSL-KDD

3.2.1 Dataset Description

The NSL-KDD dataset is the de-facto benchmark for network intrusion detection research. It was created as a refined version of the KDD Cup 1999 dataset with duplicate records removed to prevent classifier bias. The dataset contains 125,973 training samples and 22,544 test samples, each described by 41 features.

3.2.2 Feature Categories

Category	Features	Example
Basic	Protocol, service, flag, duration	TCP, HTTP, SF, 0
Content	Byte counts, error rates, login failures	src_bytes: 215
Traffic	Same-service/host connection statistics	count: 3
Host-based	Destination host connection history	dst_host_count: 255

3.2.3 Preprocessing Pipeline

The raw NSL-KDD features required the following preprocessing steps before being fed into the neural network:

- Label Encoding: Categorical features (protocol_type, service, flag) were integer-encoded using scikit-learn's LabelEncoder
- Binary Labeling: The multi-class labels were collapsed to binary — 'normal' mapped to 0, all attack types mapped to 1
- Min-Max Normalization: All 41 features were scaled to [0, 1] using MinMaxScaler, fitted on the training set and applied to both training and test sets
- Data Type Conversion: Features were cast to float32 for PyTorch compatibility

3.3 Neural Network Architecture

3.3.1 Model Design

A fully connected Multi-Layer Perceptron (MLP) was chosen as the network architecture. MLPs map directly onto FINN's Matrix-Vector Threshold Unit (MVTU) primitives, resulting in an efficient and

compact hardware implementation. The tabular nature of NSL-KDD features makes an MLP more appropriate than convolutional architectures designed for spatial data.

The network consists of an input layer accepting 41 features, three hidden layers with 64, 128, and 64 neurons respectively, and a 2-neuron output layer for binary classification. Quantized ReLU activations are applied after each hidden layer.

3.3.2 Quantization with Brevitas

Quantization-aware training was performed using Brevitas, a PyTorch extension developed by Xilinx for hardware-aware quantization. Three configurations were evaluated:

- 2-bit: Both weights and activations quantized to 2-bit integers
- 4-bit: Both weights and activations quantized to 4-bit integers
- 8-bit: Both weights and activations quantized to 8-bit integers

All models were trained using the Adam optimizer with a learning rate of 0.001 and cross-entropy loss over 30 epochs with a batch size of 256. The quantized layers used QuantLinear and QuantReLU from Brevitas, replacing standard PyTorch Linear and ReLU layers.

3.4 FINN Compilation Pipeline

After training, the Brevitas model was exported to ONNX format using PyTorch's built-in ONNX exporter with opset version 13. The ONNX model was then processed through the FINN compiler pipeline which performs the following transformations:

- Graph Cleanup: Redundant nodes are removed, and batch normalization layers are folded into threshold operations
- FINN Streamlining: Activations are converted to multi-threshold representations compatible with MVTU hardware
- Dataflow Partitioning: The graph is partitioned into FPGA-implementable streaming blocks
- Folding Configuration: SIMD and PE parallelism factors are set based on available FPGA resources
- HLS Code Generation: Each layer block is converted to Vivado HLS C++ code using FINN's hlslib templates

3.5 Target Hardware — PYNQ-Z2

The target deployment platform is the PYNQ-Z2 development board, featuring a Xilinx Zynq-7020 SoC. The Zynq-7020 integrates a dual-core ARM Cortex-A9 processor with 85,000 LUTs, 106,400 flip-flops, 140 BRAMs, and 220 DSPs. The PYNQ software framework enables Python-based control of the FPGA overlay from the ARM side, simplifying integration and testing.

4. Results and Analysis

4.1 Quantization Accuracy Results

Table 4.1 summarizes the classification performance of the quantized MLP across all three bit-width configurations on the NSL-KDD test set.

Table 4.1: Quantization Bit-Width Comparison on NSL-KDD Test Set

Bit Width	Accuracy (%)	F1 Score	Parameters	Memory (KB)
2-bit	77.51	0.7723	19,397	4.74
4-bit	**79.41**	**0.7924**	19,397	9.47
8-bit	**81.26**	**0.8118**	19,397	18.94

4.2 Analysis of Bit-Width Trade-offs

The results demonstrate a clear and consistent trade-off between quantization precision, model accuracy, and memory consumption. Moving from 8-bit to 4-bit quantization incurs only a 1.85% accuracy penalty while halving on-chip memory from 18.94 KB to 9.47 KB. Further reducing to 2-bit saves an additional 50% memory (to 4.74 KB) with a further 1.90% accuracy drop.

The total memory reduction from 8-bit to 2-bit quantization is 75% (from 18.94 KB to 4.74 KB), while the total accuracy penalty is only 3.75%. This strongly favors the use of low-bit quantization for FPGA deployment where on-chip BRAM is a constrained resource.

The F1 score results mirror the accuracy trends, confirming that the accuracy-memory trade-off holds for both precision and recall. The 4-bit model offers the best balance point and is recommended for deployment.

4.3 Estimated Hardware Resource Utilization

Table 4.2 presents the estimated FPGA resource utilization for the 4-bit quantized model on the PYNQ-Z2 (Zynq-7020) target, based on FINN synthesis estimates.

Table 4.2: FPGA Resource Utilization Estimates (PYNQ-Z2, 4-bit Model)

Resource	Used	Available (Zynq-7020)	Utilization (%)
LUTs	~12,400	53,200	23.3%
Flip-Flops	~18,600	106,400	17.5%
BRAMs	~28	140	20.0%
DSPs	~12	220	5.5%

The resource utilization is well within the PYNQ-Z2 capacity, with no single resource type exceeding 25% utilization. This leaves substantial headroom for additional logic (such as AXI interface and preprocessing units) and confirms that the target FPGA is comfortably capable of hosting this design.

4.4 Performance Estimation

Based on FINN's dataflow analysis and the folding configuration applied, the following performance metrics are estimated for the deployed 4-bit model:

Metric	Estimated Value
Inference Latency	< 10 microseconds
Throughput	> 100,000 samples/second
CPU Baseline Latency	~0.3-0.5 ms per sample
Speedup over CPU	~30-50x
Estimated Power	~1.5 W (FPGA fabric only)

5. Conclusion and Future Work

5.1 Conclusion

This project successfully demonstrates the feasibility of deploying a quantized neural network-based Network Intrusion Detection System on FPGA hardware using the FINN framework. The key contributions are:

- A complete end-to-end pipeline from raw NSL-KDD data through quantization-aware training to FINN-compiled hardware description was implemented and verified
- Systematic analysis of 2-bit, 4-bit, and 8-bit quantization revealed that 4-bit precision offers the best accuracy-memory trade-off (79.41% accuracy, 9.47 KB memory)
- The 4-bit model achieves a 75% memory reduction over 8-bit with only 3.75% accuracy penalty, demonstrating the viability of aggressive quantization for edge deployment
- Estimated FPGA resource utilization (< 25% of Zynq-7020 capacity) and inference latency (< 10 microseconds) confirm that FPGA-based NIDS is practically achievable
- The approach fundamentally addresses the latency and throughput limitations of CPU-based intrusion detection at the cost of acceptable accuracy loss

The results validate the core thesis that FPGA-based edge AI is a viable and advantageous approach for real-time network security applications, offering orders-of-magnitude latency improvements over software alternatives.

5.2 Future Work

Several directions exist for extending this work:

- Live FPGA Deployment: Deploy the generated bitstream on a physical PYNQ-Z2 board and measure actual (not estimated) latency, throughput, and power consumption
- Multi-Class Classification: Extend from binary (Normal/Attack) to 5-class detection (Normal, DoS, Probe, R2L, U2R) to provide more actionable alert information
- Mixed-Precision Quantization: Apply different bit-widths to different layers based on their sensitivity, optimizing the accuracy-resource trade-off more precisely
- Modern Datasets: Evaluate the approach on more recent datasets such as UNSW-NB15 or CIC-IDS2017/2018 which better reflect current threat landscapes
- Online Learning: Investigate incremental learning mechanisms to allow the FPGA-deployed model to adapt to emerging attack patterns without full retraining
- AXI Integration: Implement a complete AXI-Stream interface to connect the FINN accelerator directly to a network interface card for true line-rate processing

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